

1 Search Models

1.1 Exercise 7.1

A McCall Model with a chance of unemployment. The Bellman equation is as follows

$$v(w) = \max_{a \in \{\text{Accept}, \text{Reject}\}} \left\{ \frac{w}{1-\beta}, \beta(\phi(c + \bar{V}) + (1-\phi)\mathbb{E}[v(w')]) \right\} \quad (1.1)$$

where $\bar{V} = \frac{(1+\beta\phi)c + \beta(1-\phi)\mathbb{E}[v(w')]}{1-\beta\phi}$ denotes the continuation value of being unemployed.¹ $v(w)$ denotes the value function conditional on receiving an offer. $\mathbb{E}[v(w')] = \int_0^B v(w')dF(w')$ denotes the expected value conditional on receiving an offer.

1.2 Exercise 7.2

A McCall Model with two offers per period. Since $F(x) = \Pr(z \leq x)$, the cumulative distribution function of the maximum of two random variables i.i.d. drawn from $F(x)$ is $G(x_m) = \Pr(x < x_m) \cdot \Pr(x < x_m) = [F(x_m)]^2$.

$$v(w_m) = \max_{a \in \{\text{Accept}, \text{Reject}\}} \left(\frac{w_m}{1-\beta}, c + \beta\mathbb{E}_{\max}[v(w'_m)] \right) \quad (1.2)$$

where $\mathbb{E}_{\max}[v(w'_m)] = \int_0^B v(w'_m) d[F(w'_m)]^2$. Since $\mathbb{E}_{\max}[v(w'_m)] = 2 \int_0^B v(w'_m)f(w'_m)F(w'_m)dw' > \int_0^B v(w'_m)f(w'_m)dw'$,² we have

$$\bar{w}_{\max} = (1-\beta)(c + \beta\mathbb{E}_{\max}[v(w'_m)]) > (1-\beta)(c + \beta\mathbb{E}[v(w'_m)]) = \bar{w}.$$

■

1.3 Exercise 7.3

A McCall Model with n iid offers per period. Generalize the formulation in Exercise 7.2, we have

$$v(w) = \max_{a \in \{\text{Accept}, \text{Reject}\}} \left(\frac{w}{1-\beta}, c + \beta\mathbb{E}_{\max}[v(w')] \right) \quad (1.3)$$

¹I obtain $\bar{V}$ by solving the following Bellman equation: $\bar{V} = c + \beta(\phi(c + \bar{V}) + (1-\phi)\mathbb{E}[v(w')])$

²We apply the monotonicity of $v(x)$ and $2F(x) - 1$ to finish the proof:

$$\begin{aligned} & 2 \int_0^B v(w'_m)f(w'_m)F(w'_m)dw' - \int_0^B v(w'_m)f(w'_m)dw' \\ &= \int_0^B v(w'_m)(2F(w'_m) - 1)dF(w') \\ &= \mathbb{E}[v(w'_m)(2F(w'_m) - 1)] \\ &= \underbrace{\text{Cov}[v(w'_m), 2F(w'_m) - 1]}_{>0} + \underbrace{\mathbb{E}[v(w'_m)]\mathbb{E}[2F(w'_m) - 1]}_{>0} \end{aligned}$$

where $\mathbb{E}_{\max}[v(w')] = n \int_0^B v(w') F^{n-1}(w') f(w') dw'$.

1.4 Exercise 7.4

A McCall Model with a random number of offers per period. Denote the probability of n in this period but m in the next period as π_{mn} .

Part a. The Bellman Equation is given as follows.

$$v(w, n) = \max_{\{\text{Accept}, \text{Reject}\}} \left(\frac{w}{1 - \beta}, c + \beta(\mathbb{E}[v(w', n')]) \right) \quad (1.4)$$

where w is the best offer. $\mathbb{E}[v(w', n')] = \sum_{m=1}^N \pi_{mn} \int_0^B m F(w')^{m-1} v(w', n' = m) dw'$ with $\sum_m \pi_{mn} = 1$.

Part b. We characterize the reservation wage $\bar{w}(n)$ as follows.

$$\bar{w}(n) = (1 - \beta)\beta \left[\sum_{m=1}^N \pi_{mn} \int_0^B v(w', n' = m) dF(w') \right] + c(1 - \beta) \quad (1.5)$$

where $\sum_{m=1}^N \pi_{mn} \int_0^B v(w', n' = m) dF(w') \equiv \mathbb{E}[v(w', m)]$ is the marginal expected value function given the number of offers arrived.

Note that $\sum_m \pi_{mn} \mathbb{E}[v(w', m)]$ is a function of n , it follows that $\bar{w}(n)$ is also a function of n .

1.5 Exercise 7.5

A McCall Model with an endogenous number of offers per period. In this exercise we consider a case where the worker endogenously choose the number of offers $n \in 1, 2, \dots, N$ before the wages are drawn. Formally, the Bellman Equation is given by:

$$v(w_m) = \max_{a \in \{\text{Accept}, \text{Reject}\}} \left(\frac{w_m}{1 - \beta}, c + \beta \max_{n \in \{1, \dots, N\}} \left[-k(n) + \int_0^B v(w'_m) d[F(w'_m)]^n \right] \right) \quad (1.6)$$

1.6 Exercise 7.7

A McCall Model with variable labor supply. We first show that the optimal labor supply is a time invariant policy function, which solely depends on w . Let $\tilde{v}(w)$ be the value function of accepting an offer at wage w . Since the state variable w does not change across periods, the Bellman equation that characterize the optimal labor supply is given by:

$$\tilde{v}(w) = \max_{l \in [0, 1]} [u(w(1 - l), l) + \beta \tilde{v}(w)] \quad (1.7)$$

Denote the optimal labor supply given wage w by $l^*(w)$, we have

$$u(w(1 - l^*), l^*) = (1 - \beta)\tilde{v}^*(w)$$

Therefore, the optimal labor supply $l^*(w)$ does not depend on time (because $\tilde{v}^*(w)$ is time-invariant.)³

Finally, we can specify the Bellman equation using the optimal labor supply policy $l^*(w)$ as follows.

$$v(w) = \max_{\{\text{Accept}, \text{Reject}\}} \left\{ \frac{u(w(1 - l^*(w)), l^*(w))}{1 - \beta}, u(c, 1) + \beta \int_0^B v(w') dF(w') \right\} \quad (1.8)$$

Note that the preservation wage structure in the Bellman equation above is given by following equation.

$$u(w(1 - l^*(w)), l^*(w)) = (1 - \beta)(u(c, 1) + \beta \mathbb{E}[v(w')]) \quad (1.9)$$

1.7 Exercise 7.8

Wage growth rate and the reservation wage The Bellman Equation is given by

$$v(w) = \max_{\{\text{Accept}, \text{Reject}\}} \left\{ \frac{w}{1 - \beta\phi}, c + \beta \int_0^B v(w') dF(w') \right\} \quad (1.10)$$

Denote the reservation wage by $\bar{w}$, we have

$$\begin{aligned} \frac{\bar{w}}{1 - \beta\phi} &= c + \beta \left[\int_0^{\bar{w}} \frac{\bar{w}}{1 - \beta\phi} dF(w') + \int_{\bar{w}}^B \frac{w'}{1 - \beta\phi} dF(w') \right] \\ &= c + \beta \left[\frac{\bar{w}}{1 - \beta\phi} \int_0^B dF(w') + \int_{\bar{w}}^B \frac{w' - \bar{w}}{1 - \beta\phi} dF(w') \right] \end{aligned} \quad (1.11)$$

$\Rightarrow$

$$\underbrace{\frac{(1 - \beta)\bar{w}}{1 - \beta\phi}}_{g(\bar{w})} = c + \beta \underbrace{\int_{\bar{w}}^B \frac{w' - \bar{w}}{1 - \beta\phi} dF(w')}_{h(\bar{w})} \quad (1.12)$$

Define a function $G(\bar{w}) = g(\bar{w}) - h(\bar{w}) = \frac{(1 - \beta)\bar{w}}{1 - \beta\phi} - c - \frac{\beta}{1 - \beta\phi} \int_{\bar{w}}^B (w' - \bar{w}) dF(w')$. It follows that $G'(\bar{w}) = \frac{1 - \beta F(\bar{w})}{1 - \beta\phi} > 0$. Suppose $\phi_2 = \kappa\phi_1$ with $\kappa > 1$, and $G(\bar{w}_1; \phi = \phi_1) = 0$. In order to show $\bar{w}_2 < \bar{w}_1$, we need to show that $G(\bar{w}_1; \phi = \kappa\phi_1) > 0$ because $G(w)$ is increasing.

³Here I assume that the worker will stick to one optimal labor supply level under multiple equilibria.

Therefore, given the $\bar{w}_1$ corresponding to ϕ_1 , if $\beta < \frac{\bar{w}_1}{\bar{w}_1 F(\bar{w}_1) + \mathbb{E}_{w' > \bar{w}_1}[w']}$, then $\bar{w}_2 < \bar{w}_1$.

$$\begin{aligned}
G(\bar{w}_1; \phi = \phi_1) - G(\bar{w}_1; \phi = \kappa\phi_1) &= \left[\bar{w}_1(1 - \beta) - \beta \int_{\bar{w}_1}^B (w' - \bar{w}_1) dF(w') \right] \underbrace{\left(\frac{1 - \beta}{1 - \beta\phi} - \frac{1 - \beta}{1 - \beta\kappa\phi} \right)}_{\Delta\phi < 0} \\
&= \Delta\phi \left[\bar{w}_1(1 - \beta) - \beta \int_{\bar{w}_1}^B (w' - \bar{w}_1) dF(w') \right] \\
&= \Delta\phi \left[\bar{w}_1(1 - \beta) - \beta \int_{\bar{w}_1}^B w' dF(w') + \beta \int_{\bar{w}_1}^B \bar{w}_1 dF(w') \right] \\
&= \Delta\phi \left[\bar{w}_1(1 - \beta) - \beta \int_{\bar{w}_1}^B w' dF(w') + \beta \bar{w}_1(1 - F(\bar{w}_1)) \right] \\
&= \Delta\phi \left[\bar{w}_1 - \beta \underbrace{\int_{\bar{w}_1}^B w' dF(w')}_{\equiv \mathbb{E}_{w' > \bar{w}_1}[w']} - \beta \bar{w}_1 F(\bar{w}_1) \right]
\end{aligned} \tag{1.13}$$